

Domain Modeling Guidelines and Agile Modeling

Agile Modeling and Sketching

- Sketch class diagrams with open bottom and right sides to allow growth.
- Early domain models are approximate; perfection is not the goal.
- Agile models may be discarded, but can be redrawn in CASE tools if updates are desired.

Including Report Objects in a Domain Model

Receipt Example

- Reports like **Receipt** often duplicate information.
- Include only if they have a special business role (e.g., conferring return rights).
- For iteration-1 POS scenario (no returns), **Receipt** is excluded.

Use Domain Vocabulary Like a Mapmaker

- Use terms from the domain itself (e.g., library: Borrower/Patron).
- Exclude out-of-scope features (e.g., Monopoly cards not in iteration-1).
- Do not invent terms; reflect the domain accurately.

Modeling Unreal or Abstract Domains

- Some domains (e.g., telecommunications) require abstraction.
- Focus on core vocabulary from domain experts.
- Example conceptual classes: **Message**, **Connection**, **Port**, **Dialog**, **Route**, **Protocol**.

Attributes vs Conceptual Classes

- Rule of thumb: if a real-world concept is not just a number/text, it is likely a conceptual class.
- Example: **Store** is a class, not an attribute of Sale; **Airport** is a class, not an attribute of Flight.

Description Classes

Guidelines for Description Classes

- Description classes store information about other objects (e.g., `ProductDescription`, `FlightDescription`).
- Use when:
 - Information must persist even if instances are deleted (e.g., sold items).
 - Deleting instances leads to loss of critical data.
 - It reduces duplication and errors in domain information.
- Example: **`ProductDescription`** stores price, picture, description, separate from physical `Item` instances.

Service Description Example

- For services like flights or mobile packages, description classes store plan/rate details.
- Actual sold items/packages are separate instances.
- Example: **`MobileCommunicationsPackageDescription`** defines rates and features before any package is sold.